

```
CodeChef - main -
Task_3.cpp
1 /*
2  यदा यदा हि धर्मस्य ग्लानिर्भवति भारत।
3  अभ्युत्थानमधर्मस्य तदात्मानं सृजाम्यहम्॥
4  */
5  /*
6  ॐ त्र्यम्बकं यजामहे सुगन्धिं पुष्टिवर्धनम् ।
7  उर्वारुकमिव बभनान्मूल्योर्मुक्षीय मामृतात् ।।
8  */
9
10 #include<bits/stdc++.h>
11 using namespace std;
12
13 int ways(int prev, int idx, vector<int> &v1, vector<pair<int, int>> &v2, vector<vector<int>> &dp) {
14     if (idx == v1.size()) {
15         return 0;
16     }
17     if (dp[prev + 1][idx] != -1) {
18         return dp[prev + 1][idx];
19     }
20     int min_n = INT_MAX;
21     if (prev != 1 && v2[v1[idx]].second == 1) {
22         min_n = min(a: min_n, b: ways(prev: 1, idx + 1, [&] v1, [&] v2, [&] dp));
23     }
24 }
25
26 int main() {
27     int n;
28     cin >> n;
29     vector<int> v1;
30     vector<pair<int, int>> v2;
31     vector<vector<int>> dp(n, vector<int>(n, -1));
32     for (int i = 0; i < n; i++) {
33         int x;
34         cin >> x;
35         v1.push_back(x);
36     }
37     for (int i = 0; i < n; i++) {
38         int x, y;
39         cin >> x >> y;
40         v2.push_back({x, y});
41     }
42     cout << ways(1, 0, v1, v2, dp) << endl;
43     return 0;
44 }
```

Test Results 80 ms

3 tests passed 3 tests total, 80 ms

-----[Testcase #1 INPUT]-----
4
1 3 2 0

-----[Testcase #1 OUTPUT]-----
2
[+] SUCCESS: CORRECT ANSWER

-----[Testcase #2 INPUT]-----
7
1 3 3 2 1 2 3

-----[Testcase #2 OUTPUT]-----
0
[+] SUCCESS: CORRECT ANSWER

-----[Testcase #3 INPUT]-----
2
2 2

-----[Testcase #3 OUTPUT]-----
1
[+] SUCCESS: CORRECT ANSWER

```
CodeChef - main - CodeGround_4.cpp v
My parallel_algorithms.md CodeGround_4.cpp Task_3.cpp Lab_4.cpp st_algo_base.h st_tree.h st_map.h point_iterator.h st_pair.h st_vector.h Lab_3.cpp Task_1.cpp Difficulty_1.cpp
10 #include<bits/stdc++.h>
11 using namespace std;
12
13 int ways(int prev, int idx, vector<int> &v1, vector<pair<int, int>> &v2, vector<vector<int>> &dp) {
14     if (idx == v1.size()) {
15         return 0;
16     }
17     if (dp[prev + 1][idx] != -1) {
18         return dp[prev + 1][idx];
19     }
20     int min_n = INT_MAX;
21     if (prev != 1 && v2[v1[idx]].second == 1) {
22         min_n = min(a: min_n, b: ways(prev: 1, idx + 1, [&] v1, [&] v2, [&] dp));
23     }
24     if (prev != 0 && v2[v1[idx]].first == 1) {
25         min_n = min(a: min_n, b: ways(prev: 0, idx + 1, [&] v1, [&] v2, [&] dp));
26     }
27     return dp[prev + 1][idx] = min(a: min_n, b: 1 + ways(prev: -1, idx + 1, [&] v1, [&] v2, [&] dp));
28 }
29
30 int32_t main() {
31     ios_base::sync_with_stdio(false);
32     cin.tie(&tiestr: nullptr);
33 }
ways
CodeChef - CP - Task_3.cpp clang-tdy 10.0 CRLF UTF-8 4 spaces* C++ Task_3.cpp Soumyaditya Masanta <soumyaditya.masanta@gmail.com> 22:39 05-09-2025
13 int ways(int prev, int idx, vector<int> &v1, vector<pair<int, int>> &v2, vector<vector<int>> &dp) {
25     min_n = min(a: min_n, b: ways(prev: 0, idx + 1, [&] v1, [&] v2, [&] dp));
26 }
27 return dp[prev + 1][idx] = min(a: min_n, b: 1 + ways(prev: -1, idx + 1, [&] v1, [&] v2, [&] dp));
28 }
29
30 int32_t main() {
31     ios_base::sync_with_stdio(false);
32     cin.tie(&tiestr: nullptr);
33     cout.tie(&tiestr: nullptr);
34     cin.exceptions(exception: istream:: failbit);
35     int n;
36     cin >> n;
37     vector<int> v1(n);
38     vector<pair<int, int>> v2 = {{x: 0, y: 0}, {x: 0, y: 1}, {x: 1, y: 0}, {x: 1, y: 1}};
39     vector<vector<int>> dp(n: 3, value: vector<int>(n + 1, value: -1));
40     for (int i = 0; i < n; i++) {
41         cin >> v1[i];
42     }
43     cout << ways(prev: -1, idx: 0, [&] v1, [&] v2, [&] dp) << endl;
44     return 0;
45 }
```